MATH 11 — ELEMENTARY STATISTICS

§8.3

Confidence Intervals for a Population Mean

Using $\bar{x}$ to estimate μ

Today's Goal

After completing this section, you should be able to:

- Understand when to use the z-distribution vs the t-distribution
- Understand Student t-distributions and degrees of freedom
- Calculate a confidence interval for a population mean
- Interpret a confidence interval for a mean in context
- Calculate the minimum sample size needed to estimate a population mean

Learning Target 13

LT.13 Confidence Intervals

I can determine an appropriate confidence interval estimate for the proportion, mean, and standard deviation of a population and I can interpret what the confidence interval means in context.

Requirements

In order to construct a confidence interval for a population mean, the following requirements must be met.

1. The sample must be a simple random sample.
2. The population is normally distributed or $n > 30$.

When constructing a confidence interval for μ, we have two cases to consider:

1. σ is known: Use the standard normal (z) distribution.
2. σ is unknown: Use a Student t-distribution.

CI for μ (σ Known)

Procedure 8.3.1 Constructing a CI for μ with σ Known

1. Identify the values of $\bar{x}$, σ, and n.
2. Verify that the requirements above are met.
3. Identify the confidence level and find the corresponding critical value $z_{\alpha/2}$.

CI for μ (σ Known)

Procedure 8.3.1 Constructing a CI for μ with σ Known, continued

4. Find the margin of error:

$$E = z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

5. The confidence interval is

$$\bar{x} - E < \mu < \bar{x} + E.$$

The confidence interval can also be expressed in interval notation as $(\bar{x} - E,\ \bar{x} + E)$ or as $\bar{x} \pm E$.

Nightly Sleep

A simple random sample of 45 adults has a mean nightly sleep of 6.8 hours. Assume that the population standard deviation is 1.2 hours. Construct a 95% confidence interval for the mean nightly sleep for the population of adults. Round the confidence interval bounds to two decimal places.

1. We are given that $n = 45$, $\bar{x} = 6.8$, and $\sigma = 1.2$.
2. We are told that the sample is a simple random sample and $n > 30$.
3. The confidence level is 95% so $1 - \alpha = 0.95$, which means $\alpha = 0.05$. Table A-1 gives us $z_{0.025} = 1.96$.

Nightly Sleep

4. The margin of error is

$$\begin{aligned} E &= z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \\ &= 1.96 \cdot \frac{1.2}{\sqrt{45}} \\ &\approx 0.3506. \end{aligned}$$

5. The confidence interval is

$$\begin{aligned} \bar{x} - E < \mu < \bar{x} + E \\ 6.8 - 0.3506 < \mu < 6.8 + 0.3506 \\ 6.45 < \mu < 7.15. \end{aligned}$$

Interpretation: If we were to repeat this sampling process many times, 95% of the resulting confidence intervals would contain the true mean nightly sleep for the population of adults. Based on this sample, the interval $(6.45, 7.15)$ is one such interval.

Student t-Distributions

In most real-world situations, we do not know the population standard deviation σ. In this case, we use the sample standard deviation s as an estimate. However, this introduces additional uncertainty, so we cannot use the standard normal distribution. Instead, we use a Student t-distribution.

Student t-Distributions

Definition 8.3.3

If a population is normally distributed, then the distribution of the random variable

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

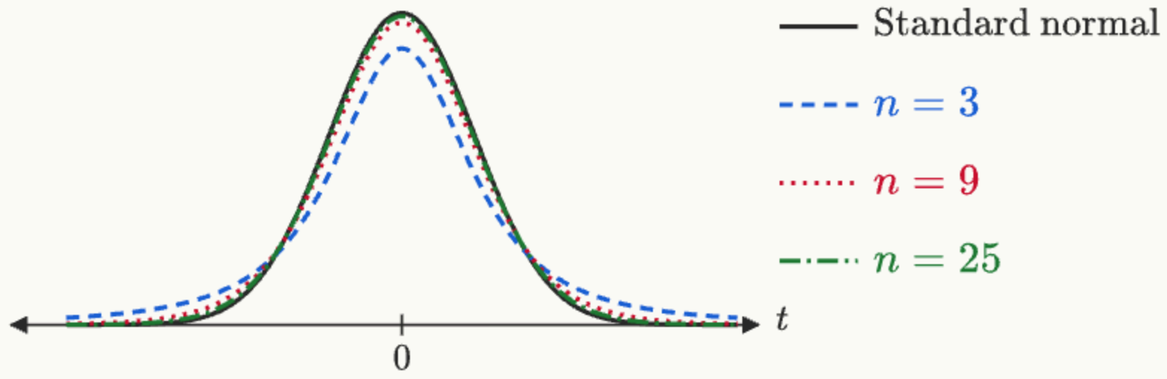
is a **Student t-distribution** for all samples of size n.

Properties

The Student t-distribution has the following properties:

- It has the same general symmetric bell shape as the standard normal distribution, but has more variability (with wider tails).
- It has a mean of $t = 0$ (just as the standard normal distribution has a mean of $z = 0$).
- The standard deviation is greater than 1 (unlike the standard normal distribution, which has $\sigma = 1$).

Changing the Sample Size



What happens to the t-distribution as n grows?

As the sample size increases, a Student t-distribution gets closer to the standard normal distribution.

Figure 8.3.4. Student t-distribution for $n = 3$, $n = 9$, and $n = 25$ compared to the standard normal distribution.

Degrees of Freedom

Definition 8.3.5

The Student t-distribution for sample size n has $n-1$ **degrees of freedom**, written as $df = n - 1$.

The degrees of freedom represent the number of values that are free to vary. When calculating the sample standard deviation, one degree of freedom is "used up" because we first calculate the sample mean $\bar{x}$.

Finding Critical t-Values

Procedure 8.3.6 Using Table A-2 for t Critical Values

To find the critical value $t_{\alpha/2}$ using Table A-2:

1. Calculate the degrees of freedom: $df = n - 1$.
2. Find the row corresponding to your degrees of freedom in the leftmost column.
3. Determine the area in one tail: $\frac{\alpha}{2}$.
4. Find the column with this area in the “Area in One Tail” row.
5. The intersection gives the critical value $t_{\alpha/2}$.

Finding Critical t-Values

Procedure 8.3.6 Using Table A-2 for t Critical Values, continued

Note: If your exact df is not in the table, use the closest df that is smaller (this gives a more conservative estimate). For large df (greater than 2000), use the "Large" row.

Using Excel for t Critical Values

Procedure 8.3.7 Using Excel for t Critical Values

To find the critical value $t_{\alpha/2}$ for a confidence interval, use the `T.INV.2T` function:

```
=T.INV.2T(alpha, df)
```

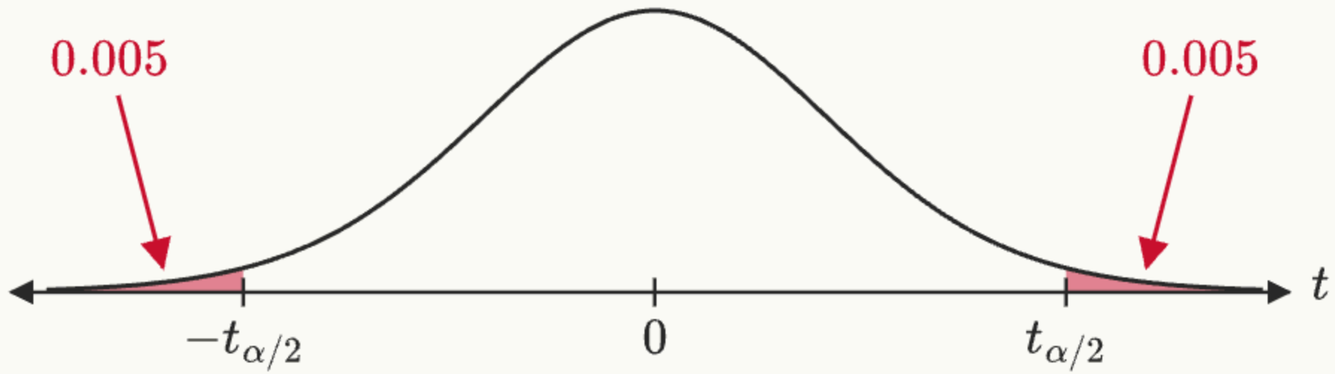
- `alpha` = the significance level (e.g., 0.05 for 95% confidence).
- `df` = degrees of freedom ($n - 1$).

Example: For 95% confidence with $n = 15$: `=T.INV.2T(0.05, 14)` returns 2.145.

Finding a Critical Value

Find the critical value $t_{\alpha/2}$ corresponding to a 99% confidence level, given that the sample has size $n = 20$.

The confidence level is 99% so $1 - \alpha = 0.99$, which means $\alpha = 0.01$ is the total area in two tails. So $\frac{\alpha}{2} = 0.005$ is the area in each tail.



Finding a Critical Value

Since $n = 20$, we have $df = n - 1 = 19$ degrees of freedom.

Using Table A-2:

1. Find $df = 19$ in the "Degrees of Freedom" column on the left.
2. Find 0.005 in the "Area in One Tail" row at the top.
3. The intersection of this row and column gives $t_{0.005} = 2.861$.

Using Excel, we would enter `=T.INV.2T(0.01, 19)` into any empty cell. This returns a more accurate value of $t \approx 2.8609$.

CI for μ (σ Unknown)

Procedure 8.3.9 Constructing a CI for μ with σ Unknown

1. Identify the values of $\bar{x}$, s, and n.
2. Verify that the requirements above are met.
3. Identify the confidence level and find the corresponding critical value $t_{\alpha/2}$ with $df = n - 1$.
4. Find the margin of error:

$$E = t_{\alpha/2} \cdot \frac{s}{\sqrt{n}}$$

CI for μ (σ Unknown)

Procedure 8.3.9 Constructing a CI for μ with σ Unknown, continued

5. The confidence interval is

$$\bar{x} - E < \mu < \bar{x} + E.$$

The confidence interval can also be expressed in interval notation as $(\bar{x} - E,\ \bar{x} + E)$ or as $\bar{x} \pm E$.

Notice that this procedure is the same as when σ is known, except that we use $t_{\alpha/2}$ for the critical value instead of $z_{\alpha/2}$ and we use s in place of σ.

Which Distribution?

Each sample below is used to estimate a population mean μ.

(a) A simple random sample of 50 adults, with σ known.

σ is known: use the standard normal (z) distribution.

(b) A simple random sample of 50 adults, with only s known.

σ is unknown: use a Student t-distribution.

(c) A simple random sample of 12 adults from a population that is strongly skewed to the right.

Neither. The population is not normally distributed and n is not greater than 30, so the requirements are not met.

Weekly Homework Time

In a simple random sample of 44 college students, the mean time spent on homework was 12.6 hours per week and the standard deviation was 4.8 hours. Construct a 95% confidence interval for the mean weekly homework time of college students. Round the confidence interval bounds to one decimal place.

1. We are given that $n = 44$, $\bar{x} = 12.6$, and $s = 4.8$.
2. We are told that the sample is a simple random sample and $n > 30$.

Weekly Homework Time

3. The confidence level is 95% so $1 - \alpha = 0.95$, which means $\alpha = 0.05$. So $\frac{\alpha}{2} = 0.025$. With $df = n - 1 = 43$ degrees of freedom, we use Table A-2. Since $df = 43$ is not listed, we use $df = 40$ (rounding down for a more conservative estimate). Looking at the row for $df = 40$ and the column for Area in One Tail $= 0.025$, we get $t_{0.025} = 2.021$.

Weekly Homework Time

4. The margin of error is

$$\begin{aligned} E &= t_{\alpha/2} \cdot \frac{s}{\sqrt{n}} \\ &= 2.021 \cdot \frac{4.8}{\sqrt{44}} \\ &\approx 1.4625. \end{aligned}$$

5. The confidence interval is

$$\begin{aligned} \bar{x} - E < \mu < \bar{x} + E \\ 12.6 - 1.4625 < \mu < 12.6 + 1.4625 \\ 11.1 < \mu < 14.1. \end{aligned}$$

Interpretation: If we were to repeat this sampling process many times, 95% of the resulting confidence intervals would contain the true mean weekly homework time of college students. Based on this sample, the interval $(11.1, 14.1)$ is one such interval.

Phone Battery Life

A simple random sample of 25 phones of one model has a mean battery life of 14.2 hours and a standard deviation of 2.5 hours. Battery life for this model is normally distributed. Construct a 90% confidence interval for the mean battery life of all phones of this model. Round the confidence interval bounds to two decimal places.

Work this one with a partner.

Phone Battery Life

1. We are given that $n = 25$, $\bar{x} = 14.2$, and $s = 2.5$.
2. n is not greater than 30, but we are told that the population is normally distributed.
3. $\alpha = 0.10$, so $\frac{\alpha}{2} = 0.05$. With $df = 24$, Table A-2 gives $t_{0.05} = 1.711$.
4. $E = 1.711 \cdot \frac{2.5}{\sqrt{25}} = 0.8555$
5. $14.2 - 0.8555 < \mu < 14.2 + 0.8555$, so $13.34 < \mu < 15.06$.

Sample Size for Estimating a Mean

Before collecting data, we often need to decide how large a sample to draw. Inverting the margin-of-error formula for a mean lets us calculate the minimum sample size required to achieve a desired margin of error.

Procedure 8.3.11 Sample Size for Estimating a Mean

To determine the sample size required to estimate a population mean with a specified margin of error E and confidence level $1 - \alpha$, we use:

$$n = \left(\frac{z_{\alpha/2} \cdot \sigma}{E} \right)^2$$

Important: Always round the result **up** to the nearest whole number.

Daily Exercise

We want to construct a 95% confidence interval for the mean daily exercise time of adults. A pilot study gives a standard deviation of 12 minutes, which we use for σ. If we want the population mean to be within 3 minutes of the sample mean, how big does the sample need to be? Round up to the nearest whole number.

The confidence level is 95% so $1 - \alpha = 0.95$, which means that $\alpha = 0.05$. Table A-1 gives us $z_{0.025} = 1.96$. We are given that the margin of error is $E = 3$.

$$n = \left(\frac{z_{\alpha/2} \cdot \sigma}{E} \right)^2 = \left(\frac{(1.96)(12)}{3} \right)^2 \approx 61.47$$

Rounding up, we need a sample of size $\mathbf{62}$.

Why z and Not t?

Note that this formula requires knowing (or estimating) the population standard deviation σ. In practice, we often use the sample standard deviation s from a pilot study or previous research as an estimate for σ.

When σ is unknown, the confidence interval uses $t_{\alpha/2}$. The sample size formula uses $z_{\alpha/2}$.

Why can't the sample size formula use $t_{\alpha/2}$?

Finding $t_{\alpha/2}$ requires $df = n - 1$, and n is exactly what we are solving for, so we plan with $z_{\alpha/2}$ and accept that the interval may turn out very slightly wider than planned.